THE COMPLETE TOUR

# JanVayu, end to end

Every section of the platform, one slide at a time: the live data, the health toll, the money, accountability and the tools.



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CHAPTER 01

# Why it exists

01



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WHY IT EXISTS

# India breathes the world's most dangerous air

- **1.72 million** deaths a year from ambient PM2.5 (Lancet Countdown 2025)
- **3.5 years** of life expectancy lost on average (AQLI 2025)
- **6th** most-polluted country; Loni the most-polluted city on Earth (IQAir 2025)
- India and China each recorded more than 2 million air-pollution deaths in 2023 (State of Global Air 2025)

The scale is why JanVayu exists: to turn scattered official data into a public record anyone can act on.



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WHY IT EXISTS

# A citizen-led public record

- **No party line**: we track promises and money, and endorse no party
- **Citizen-led and open**: anyone can read, reuse or improve it
- **Every number cites its source**, and is periodically re-verified against primary sources
- Free, forever, no ads

The case for clean air is strong enough that it never needs an exaggerated number.



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WHY IT EXISTS

# One clear picture, from many sources

- **CPCB / CAAQMS**: India’s official monitoring network
- **WAQI, IQAir, OpenAQ**: global aggregators and community sensors
- **CREA, WHO, Lancet, NASA**: research and satellite data
- JanVayu cleans and cross-checks them into one picture you can trust

Scattered official data, turned into something a citizen can actually read.



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CHAPTER 02

# The live picture

02



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THE LIVE PICTURE

# The dashboard

- Live **PM2.5** for your city in µg/m³, with the AQI beside it and a plain-language health band
- Cigarette-equivalent and disease-risk read-outs
- “Did you know” sourced facts and an anomaly explainer
- Everything one tap from the city selector at the top

The home screen answers “how bad is it right now, and what does that mean for me?”



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THE LIVE PICTURE

# Real-time PM2.5 for 157 Indian cities

- 157 Indian cities (and three abroad for comparison), ranked and coloured on **PM2.5**: the fraction that reaches the bloodstream
- Type a name to jump to any city
- Covers the major NCAP non-attainment towns
- Every value shown with its source and time

The whole country is one city tap away, and every headline number is PM2.5, not an index.



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THE LIVE PICTURE

# The live map

- Every monitoring station plotted, colour-coded by AQI
- **MPs / MLAs / Districts**: all 543 Lok Sabha seats and 785 districts coloured by live air; grey where no monitor is near
- **Sources**: landfills, dumpsites, coal mines, red/orange industrial parks, SEZs
- Constituency pop-ups link straight to the Accountability tracker and RTI templates

Where the numbers come from, and the air your MP answers for.



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THE LIVE PICTURE

# Rankings & comparison

- **City Rankings**: cleanest to worst, live / 7-day / 30-day
- **Compare**: any two cities side by side, with a verdict
- International benchmarks (WHO guideline, world cities)
- Shareable cards for any ranking or comparison

Context turns a number into a story.



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THE LIVE PICTURE

# Trends & forecast

- **Trends**: scrubbable 24-hour and year-over-year charts
- **Forecast**: a free 5-day PM2.5 outlook (Open-Meteo / CAMS)
- **Farm Fire Tracker**: live stubble fires (NASA FIRMS) across the Punjab–Haryana–NCR belt
- Ask JanVayu can answer “will it be bad tomorrow?”

Look back to understand, look forward to plan.



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THE LIVE PICTURE

# One map, every place in India

- A **Boundaries** menu covering **983,149 areas across seven levels**: state, district, block/mandal/tehsil, gram panchayat and village; city and ward
- A **Colour** menu with nine measures: annual PM2.5, the same air by season, surface heat, tree cover, green cover and built-up
- Every area carries all nine, with no dropdown and no list of supported cities
- A tenth, **this year so far**, answers the question everyone asks about the 2024 date. It covers districts only, because it is a ~40 km model corrected against the satellite series, and it rebuilds every month
- **Compare**: plot any two measures against each other for whatever is on screen
- **Who breathes it**: schools and health centres over any level

You should not have to know whether your home is officially a ward or a village to find it.



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THE LIVE PICTURE

# Every village in India

- All **584,615** village boundaries, one of the seven levels in the same menu
- Each carries the full set: annual and seasonal PM2.5, surface heat, tree, green and built-up cover
- Coloured against the WHO guideline (5) and India’s limit (40)
- The live estimate stays separate, and says so when no monitor is near

Two-thirds of India doesn’t live in a city. Now it’s on the map.



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THE LIVE PICTURE

# Your airshed, or your town?

- Across **785 districts**, **89.2%** of the variance in annual PM2.5 sits *between* states, not within them
- State medians run Delhi **92.7** down to Ladakh **13.9**; 13 states and UTs are above India's own limit of 40
- Pick a district and the panel splits its distance from the national median into region and local deviation
- **43 years** of PM2.5 underneath it: 783 districts, 1980 to 2022, annual and by season

NCAP sets targets city by city. Most of the burden arrives from the airshed.



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THE LIVE PICTURE

# Alerts, correlations & the pollution calendar

- **PM2.5 alerts**: Web Push when your city crosses your threshold
- **Pollution Correlations**: how AQI moves with weather and events
- **Pollution Calendar**: the seasonal shape of your city’s air
- **Installable app**: add JanVayu to your home screen; alerts arrive even when it’s closed

Turn monitoring into something that reaches you.



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CHAPTER 03

# Health & exposure

03



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HEALTH & EXPOSURE

# What the air does to a body

- **Health Impact**: a simple log-concentration mortality risk score for your exposure (not GEMM)
- **Beyond the Lungs**: heart, brain, kidneys, metabolism, pregnancy
- Anchored on recent Indian cohort studies, cited in-line
- Translates a µg/m³ number into a health consequence

PM2.5 harms far more than the lungs.



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HEALTH & EXPOSURE

# The most vulnerable

- **Children's Health**: smaller lungs, higher dose, developing brains
- **Women's Health**: cooking-smoke exposure, maternal & pregnancy risk
- Household air pollution and the clean-fuel gap
- Each with sourced figures and action steps

Exposure is not shared equally.



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HEALTH & EXPOSURE

# Your own exposure

- **Exposure Report**: your estimated daily dose
- **Exposure Diary**: 16 activities with illustrative PM2.5 multipliers
- **Indoor Air**: why inside is often worse, plus a sensor buying guide
- Cigarette-equivalent and life-expectancy read-outs

Make the invisible personal.



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HEALTH & EXPOSURE

# What to do about it

- **Should I Go Outside?**: a today-specific yes/no by activity
- **Purifier Calculator**: the right CADR for your room
- **School Closures**: the GRAP stages that move classes to hybrid mode (and the closures states may add)
- **Occupational Exposure**: vendors, traffic police, gig riders, waste workers

From “how bad is it” to “what do I do today”.



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CHAPTER 04

# The economic bill

04



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THE ECONOMIC BILL

# The bill nobody sees

- **Economic Cost**: India’s ~US$339.4B (9.5% of GDP): the monetised value of 2022 premature deaths from outdoor PM2.5 (Lancet Countdown 2025)
- Pandey et al. 2021 put lost output at US$36.8B in 2019, a different method
- **Migration & Displacement**: when air drives people to move
- **Migration Calculator**: life-years gained by relocating

Dirty air costs the economy as well as people’s health.



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CHAPTER 05

# Following the money

05



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FOLLOWING THE MONEY

# Policy & budgets

- **Policy Tracker**: NCAP, GRAP, court orders and their status
- **Budget Tracker**: where clean-air money is allocated vs spent
- **City Policy Tracker**: targets and expenditure city by city
- The gap between promise and delivery, made visible

Accountability starts with following the money.



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FOLLOWING THE MONEY

# Scorecards & the mission tracker

- **City Scorecards**: a graded report card per city, with one-click RTI
- **Mission Tracker**: independent E1–E5 verification of NCAP claims
- **Political Accountability**: promises vs reality, dated
- **Accountability Brief**: an auto-generated ward-level brief

Turn data into pressure.



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FOLLOWING THE MONEY

# Sources, wins & the law

- **Industrial Sources**: the plants and sectors behind the smog
- **Where PM2.5 comes from**: the per-city source-apportionment ring
- **Clean Air Wins**: what has actually worked
- **Legal Framework**: your rights, key rulings, and how to use them

Name the sources; know your rights.



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FOLLOWING THE MONEY

# The RTI Assistant

- Pick a state and a topic; it drafts a filing-ready RTI
- Pre-filled from a scorecard when a city misses its target
- Names the right pollution-control authority
- Turns a citizen’s frustration into a formal question

The single most actionable tool on the site.



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CHAPTER 06

# Citizen action

06



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CITIZEN ACTION

# Act, learn, be heard

- **Take Action / Citizen Plan**: concrete steps for your situation
- **Voices & Field Testimony**: the social record and on-ground accounts
- **Workshops**: book a guided session, plus a 10-question air-literacy self-check
- **Learning Games**: seven original India-context games

A platform you use, not just read.



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CITIZEN ACTION

# News, in context

- **Social Feed**: Reddit, YouTube and Indian news, refreshed automatically
- **Live News**: the latest reporting, aggregated
- **Photo Gallery**: “The air, in pictures”, 31 documentary photographs
- **From the blog**: the newest post and the three behind it, on the dashboard

Keep the issue alive between crises.



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CHAPTER 07

# Tools & knowledge

07



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TOOLS & KNOWLEDGE

# Ask JanVayu

- A chatbot that answers about your city and ward
- Seasonal, forecast and NCAP context built in
- Answers in ten languages, with sources
- Your fastest way into the data

Natural-language access to everything above.



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TOOLS & KNOWLEDGE

# Learn & look things up

- **Understanding AQI**: what the number actually rolls up
- **Glossary**: the vocabulary, one keystroke away (Ctrl-K)
- **Reading List**: 66 peer-reviewed studies, downloadable as BibTeX or RIS
- **FAQ**: 30 sourced answers with a live search

From curious to informed.



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TOOLS & KNOWLEDGE

# Take the data with you

- **Downloads**: briefs, datasets, and citation-ready material
- **Data Archive**: a growing record of readings over time
- **Open Data API**: pull the data straight into your own work, no login
- **Embeddable widgets**: drop live AQI or rankings into any site

Built for researchers, journalists and advocates.



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CHAPTER 08

# Trust

08



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TRUST

# Every number, checked against its source

- Every figure the site states is recomputed from the data on **every change**, and a mismatch stops the release
- A claim we have **retracted** cannot come back; the site refuses it in five languages
- A periodic deep audit re-verifies statistics against primary sources; findings are logged publicly
- Two honest methods are never collapsed into one, and unverifiable numbers are flagged rather than guessed

Honesty has to hold up behind the scenes too.



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TRUST

# We check our own map against the monitors

- Every PM2.5 figure on the site is **modelled** (satellite, reanalysis or reconstruction), and each says so
- There is also a **measured** layer: CPCB's own hourly record, via the XKDR India Air Quality Database (CC BY 4.0)
- For 2024 the satellite map and the station readings agree at **r = 0.834**, the satellite about 2 µg/m³ lower
- That gap is the right direction, not an error, because a district average includes the fields around the city

A model you never compare against an instrument is a claim, not a measurement.



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TRUST

# Eleven years of the government’s own bulletin

- CPCB publishes a daily AQI bulletin as a PDF, and has since 1 May 2015. It was public, and effectively unreadable as a series
- We hold it as one table: **300 cities, 2015 to 2026**, 517,157 city-days, days in each official category per city per year
- Beside every figure is **the number of monitors behind it**, which is the part that changes what the number means
- In CPCB’s own 2024 bulletin, **204 of 264 cities had a citywide AQI from a median of exactly one monitor**

The record was always public. It was not usable.



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TRUST

# Was it policy, or was it the wind?

- A city's air can improve because it emitted less, or because the wind blew harder
- We separate the two for **44 cities, 2018–2024**, using random forests over local weather (Grange et al. 2018)
- **33 are improving** once weather is removed. **11 are not.**
- Steepest falls: Meerut −14.6, Varanasi −14.2, Lucknow −14.0 µg/m³ a year. Rising: Chandigarh +3.1, Gwalior +2.4, Mumbai +0.8

Every "air improved by X%" headline rests on not knowing which of the two happened.



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TRUST

# Live against annual

- A single day’s AQI swings with the weather
- Annual averages are the fair yardstick for comparison
- Ward maps show a live SPREAD, not a per-street truth
- We say so, everywhere the distinction matters

Being right includes being clear about uncertainty.



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GET INVOLVED

# Clean air is everyone’s right.

The platform is open source and the data is yours. Pull it, embed it, file an RTI with it, or add your name to the team at github.com/JanVayu/JanVayu.



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